

Clinical Pathology Residency

High-Yield Lecture Curriculum

*A 34-lecture, case-based program across seven clinical pathology domains,
built on the department's own teaching repository and current professional standards.*

Prepared for

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About This Curriculum

This curriculum organizes clinical pathology didactics into **seven subspecialty blocks and 34 high-yield lectures**, each designed as a 45–60 minute, case-based session. The emphasis is the work residents actually do on service: test selection and interpretation, laboratory stewardship, troubleshooting, and consultative practice — not exhaustive theory.

HOW IT IS BUILT

Every lecture is mapped to source material already in the department's teaching repository — the same indexed library of 1,600+ files catalogued in the Clinical Pathology dashboard. Each lecture specification below lists the specific repository documents (resident lectures, reference-text chapters, rotation handouts) that supply its substance. Where the repository is thin on a topic (for example, parts of immunology/serology and molecular variant interpretation), the content is completed from current professional standards and flagged as a gap to develop locally.

CONTENT STANDARDS

Material is reconciled against current authorities including the AABB Technical Manual and Standards, CLSI guidelines, CAP/CLIA requirements, WHO classifications, and consensus definitions (e.g., the 2019 TRALI redefinition and 2018 ISBT TACO criteria). All diagrams and algorithms in the slide decks are original and drawn for this series.

REPEATED LECTURE FRAMEWORK

Each block follows the same arc — **fundamentals** → **methodology** → **interpretation** → **pitfalls** → **case application** — so residents move from memorization toward consultative judgment.

Each session is structured as: **10 min** core concepts · **20 min** interpretive framework or algorithm · **15 min** real cases with pitfalls · **5–10 min** board-style review.

Curriculum at a Glance

Thirty-four sessions across seven blocks. Each lecture has a matching slide deck (filename shown).

HEMATOLOGY & COAGULATION · 5 lectures		
1	CBC and Peripheral Smear Interpretation	Heme_01.pptx
2	Anemia Workup and Red Cell Disorders	Heme_02.pptx
3	Leukocytosis, Cytopenias, and Acute Leukemia Screening	Heme_03.pptx
4	Platelet Disorders and Thrombocytopenia	Heme_04.pptx
5	Coagulation Testing and Bleeding / Thrombosis Evaluation	Coag_05.pptx
TRANSFUSION MEDICINE · 5 lectures		
6	ABO/Rh, Antibody Screen, and Compatibility Testing	Trans_01.pptx
7	Antibody Identification and Complex Serology	Trans_02.pptx
8	Transfusion Reactions	Trans_03.pptx
9	Component Therapy and Massive Transfusion	Trans_04.pptx
10	Special Topics in Blood Bank	Trans_05.pptx
CLINICAL CHEMISTRY · 5 lectures		
11	Analytic Techniques and Preanalytic / Analytic Error	Chem_01.pptx
12	Electrolytes, Osmolality, and Acid–Base Interpretation	Chem_02.pptx
13	Renal, Liver, and Cardiac Biomarkers	Chem_03.pptx
14	Endocrine and Protein Testing	Chem_04.pptx
15	Toxicology, TDM, and Point-of-Care Testing	Chem_05.pptx
MICROBIOLOGY · 5 lectures		
16	Specimen Selection, Collection, and Rejection Criteria	Micro_01.pptx
17	Gram Stain, Culture Workflows, and Organism Identification	Micro_02.pptx
18	Antimicrobial Susceptibility Testing and Reporting	Micro_03.pptx
19	Rapid Diagnostics and Molecular Microbiology	Micro_04.pptx
20	Blood Cultures, Sepsis, and Infection-Control Pearls	Micro_05.pptx
IMMUNOLOGY & SEROLOGY · 4 lectures		
21	Immunoassay Principles and Test Performance	Immuno_01.pptx
22	Autoimmune Serology	Immuno_02.pptx
23	Infectious Disease Serology	Immuno_03.pptx
24	Allergy, Immunodeficiency, and Protein-Based Immune Testing	Immuno_04.pptx
MOLECULAR & CYTOGENETICS · 5 lectures		

25	Molecular Methods for Residents	Molecular_01.pptx
26	Variant Interpretation and Reporting	Molecular_02.pptx
27	Molecular Testing in Hematologic Malignancy	Molecular_03.pptx
28	Solid Tumor and Germline Testing Fundamentals	Molecular_04.pptx
29	Validation, Quality Control, and Bioinformatics Essentials	Molecular_05.pptx
LABORATORY MANAGEMENT, INFORMATICS, QUALITY & SAFETY · 5 lectures		
30	Quality Management Systems	LabMgmt_01.pptx
31	Regulatory and Accreditation Essentials	LabMgmt_02.pptx
32	Patient Safety and Error Investigation	LabMgmt_03.pptx
33	Laboratory Informatics and LIS Fundamentals	LabMgmt_04.pptx
34	Utilization Management, Cost Awareness, and Leadership	LabMgmt_05.pptx

Block 1 · Hematology & Coagulation

5 lectures

1. CBC and Peripheral Smear Interpretation

45–60 min · deck: *Heme_01.pptx*

LEARNING OBJECTIVES

- Interpret CBC indices and automated analyzer flags in clinical context.
- State the triggers for which a manual smear review changes management.
- Recognize high-yield red cell, white cell, and platelet morphologies.
- Identify spurious results and their preanalytic causes.
- Communicate urgent morphologic findings to the clinical team.

KEY TOPICS

- RBC indices (MCV, MCH, RDW) and reticulocytes
- WBC differential and analyzer flags
- Platelet estimate and clumping
- Smear-review triggers and laboratory rules
- Key morphologies: schistocytes, spherocytes, blasts, target cells
- Critical-result morphology and reporting

BOARD-EXAM THEMES

- Pseudo-abnormalities: platelet clumping, cold agglutinins, lipemia
- Blast recognition and when to escalate
- Schistocyte significance and the microangiopathy clue

SOURCE MATERIALS IN THE REPOSITORY

- A Peripheral Blood Smear (*Hematopathology / Lectures*)
- Peripheral blood cell morphology (*Hematopathology / Lectures*)
- Heme CP on call conference Peripheral blood smear review in malignancy (*Hematopathology*)
- Barth peripheral smear analysis (*Hematopathology*)
- hematology basic investigations on call CP role (*Hematopathology*)
- heme reference ranges (*Hematopathology*)

2. Anemia Workup and Red Cell Disorders

45–60 min · deck: *Heme_02.pptx*

LEARNING OBJECTIVES

- Apply an MCV-based algorithm to micro-, normo-, and macrocytic anemia.
- Interpret iron studies and distinguish IDA, ACD, and thalassemia trait.
- Construct a laboratory workup for hemolysis.
- Recognize hemoglobinopathy clues on indices and smear.
- Avoid common interpretive pitfalls (e.g., reticulocyte response).

KEY TOPICS

- MCV-based diagnostic algorithm
- Iron studies: ferritin, transferrin saturation, sTfR

- Reticulocyte count and production index
- Hemolysis panel: LDH, haptoglobin, bilirubin, DAT
- B12 / folate and macrocytosis
- Hemoglobinopathy screening overview

BOARD-EXAM THEMES

- IDA vs ACD vs thalassemia trait (Mentzer, RDW, ferritin)
- Intravascular vs extravascular hemolysis labs
- Interpreting an inappropriately low reticulocyte count

SOURCE MATERIALS IN THE REPOSITORY

- Warm Autoimmune Hemolytic Anemia HM 02262020 (*Hematopathology*)
- Heme Hemoglobinopathy Evaluation Keren (*Hematopathology*)
- Heme Hemoglobins kezlarian (*Hematopathology*)
- Unraveling Hemoglobinopathies Keren (*Hematopathology*)
- TMH S2 Hemolytic Uremic Syndrome (*Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis*)
- Harmening 20 Autoimmune Hemolytic Anemia (*Transfusion Medicine / Harmening*)

3. Leukocytosis, Cytopenias, and Acute Leukemia Screening

45–60 min · deck: *Heme_03.pptx*

LEARNING OBJECTIVES

- Evaluate leukocytosis and leukopenia with a structured differential.
- Recognize urgent findings that demand same-day action.
- Identify blasts and Auer rods on the smear.
- Know when to escalate to flow cytometry and marrow examination.
- Distinguish leukemoid reaction, CML, and acute leukemia.

KEY TOPICS

- Differential of leukocytosis and left shift
- Pancytopenia: production vs destruction
- Blast morphology and Auer rods
- Acute promyelocytic leukemia as an emergency (DIC)
- Reactive vs neoplastic lymphocytosis
- Triggers for flow cytometry / marrow

BOARD-EXAM THEMES

- APL recognition and the coagulopathy alert
- Leukemoid reaction vs CML (LAP, BCR-ABL1)
- Estimating blast percentage and reporting urgency

SOURCE MATERIALS IN THE REPOSITORY

- Heme Lymphoid Neoplasms (*Hematopathology*)
- Bone Marrow Part I (*Hematopathology / Lectures*)
- Bone Marrow part 2 (*Hematopathology / Lectures*)
- AML questions session with Dr Ghosh (*Hematopathology / Lectures*)

- Heme FlowCytometry Intro v (Hematopathology / Flow panel descriptions)
- Dr Carey's Intro To Clinical Flow Cytometry (Hematopathology / Lectures)

4. Platelet Disorders and Thrombocytopenia

45–60 min · deck: *Heme_04.pptx*

LEARNING OBJECTIVES

- Differentiate pseudothrombocytopenia from true thrombocytopenia.
- Classify thrombocytopenia by mechanism.
- Recognize TTP, HIT, ITP, and DIC by their laboratory signatures.
- State the transfusion implications of each cause.
- Outline platelet function testing.

KEY TOPICS

- EDTA-induced platelet clumping
- Production vs destruction vs sequestration
- ITP as a diagnosis of exclusion
- TTP: schistocytes and ADAMTS13
- Heparin-induced thrombocytopenia (HIT)
- DIC laboratory profile; platelet function assays

BOARD-EXAM THEMES

- Pseudothrombocytopenia and the citrate-tube confirmation
- TTP pentad and ADAMTS13 testing
- HIT: 4T score, timing, and the paradox of thrombosis

SOURCE MATERIALS IN THE REPOSITORY

- Coag Chapter 45 Thrombocytopenia in the Newborn Roberts Platelets (Coagulation)
- TMH S2 Platelet Function Analyzer (Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis)
- TMH S2 Congenital Thrombocytopenia (Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis)
- TMH S2 Platelet Aggregation Studies (Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis)
- TMH S2 Drug Induced Thrombocytopenia (Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis)
- TMH S2 Antiplatelet Therapy Monitoring (Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis)

5. Coagulation Testing and Bleeding / Thrombosis Evaluation

45–60 min · deck: *Coag_05.pptx*

LEARNING OBJECTIVES

- Interpret PT, aPTT, fibrinogen, and D-dimer.
- Use mixing studies to separate factor deficiency from inhibitor.
- Outline lupus anticoagulant testing.
- Build a rational thrombophilia testing strategy and timing.
- Recognize preanalytic errors unique to coagulation.

KEY TOPICS

- PT / aPTT pathways and common causes of prolongation
- Mixing studies: correction vs non-correction
- Factor assays and specific inhibitors
- Lupus anticoagulant / antiphospholipid testing
- Thrombophilia panel and when (not) to test
- Preanalytic pitfalls: tube fill, heparin contamination

BOARD-EXAM THEMES

- Isolated prolonged aPTT differential
- Mixing study interpretation
- Timing of thrombophilia testing relative to event and anticoagulation

SOURCE MATERIALS IN THE REPOSITORY

- PRINCIPLES OF CLOTTING FACTOR THERAPY IN HEMOPHILIA A and B (*Coagulation*)
- COMPREHENSIVE OVERVIEW Inhibitors in Hemophilia A and B (*Coagulation*)
- TMH S2 Hemophilia B (*Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis*)
- TMH S2 Hemophilia A (*Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis*)
- TMH S2 Mixing Studies (*Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis*)
- TMH S2 Factor XI Deficiency (*Coagulation / Transfusion Medicine and Hemostasis (Book) - Section 2 Hemostasis*)

Block 2 · Transfusion Medicine

5 lectures

6. ABO/Rh, Antibody Screen, and Compatibility Testing

45–60 min · deck: *Trans_01.pptx*

LEARNING OBJECTIVES

- Explain ABO and Rh typing and resolve common discrepancies.
- Describe the antibody screen and its role in pretransfusion testing.
- Contrast immediate-spin and antiglobulin crossmatch.
- Apply electronic crossmatch criteria.
- State emergency-release principles.

KEY TOPICS

- ABO forward/reverse typing and discrepancies
- Rh(D) typing, weak D, partial D
- Antibody screen and pretransfusion workflow
- Immediate-spin vs AHG crossmatch
- Electronic / computer crossmatch
- Emergency release and group O use

BOARD-EXAM THEMES

- ABO discrepancy resolution
- When the immediate-spin crossmatch is sufficient
- Emergency-release documentation

SOURCE MATERIALS IN THE REPOSITORY

- Harmening 7 The Rh Blood Group System (*Transfusion Medicine / Harmening*)
- Harmening 6 The ABO Blood Group System (*Transfusion Medicine / Harmening*)
- TMH S1 C H A P T E R 2 3 ABO and H Blood Group System (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- TMH S1 C H A P T E R 2 4 Rh and RhAG Blood Group Systems (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- TMH S1 C H A P T E R 2 2 Molecular DNA Based Blood Group Typing (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- ABOHL (*Transfusion Medicine / Dr. Brancamp Lectures*)

7. Antibody Identification and Complex Serology

45–60 min · deck: *Trans_02.pptx*

LEARNING OBJECTIVES

- Work an antibody identification panel systematically.
- Recognize common clinically significant alloantibodies.
- Approach warm autoantibodies and the adsorption concept.
- Identify high-yield problem-solving patterns (dosage, enzymes).
- Select antigen-negative units appropriately.

KEY TOPICS

- Panel interpretation: ruling out and in
- Clinically significant alloantibodies (Rh, Kell, Kidd, Duffy)
- Warm autoantibody workup; adsorptions
- Enzyme effects and antigen dosage
- Phenotyping and antigen-negative selection
- High-frequency / low-frequency antigens

BOARD-EXAM THEMES

- Single vs multiple alloantibodies on a panel
- Anti-Jk^a dosage and evanescence
- Warm autoantibody masking an alloantibody

SOURCE MATERIALS IN THE REPOSITORY

- Pretransfusion testing and antibody identification lectures (*Transfusion Medicine / Dr Otrrock lectures*)
- TMH S1 C H A P T E R 2 0 Antibody Identification (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- TMH S1 C H A P T E R 2 5 Kell and Kidd Blood Group Systems (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- DAT and Immune Hemolysis (*Transfusion Medicine / Dr. Eisenbrey Lectures*)
- Harmening 5 The Antiglobulin Test (*Transfusion Medicine / Harmening*)
- AABB Chap17 Direct Antiglobulin Test (*Transfusion Medicine / AABB Technical Manual*)

8. Transfusion Reactions

45–60 min · deck: *Trans_03.pptx*

LEARNING OBJECTIVES

- Execute the universal first response to a suspected acute reaction.
- Classify reactions by timing and mechanism.
- Manage hemolytic, febrile, allergic/anaphylactic, and septic reactions.
- Distinguish TACO from TRALI.
- Select product modifications to prevent recurrence.

KEY TOPICS

- First response: stop–stabilize–verify–notify
- Acute hemolytic (AHTR) and the ABO error
- FNHTR, allergic, and anaphylactic reactions
- Septic reactions and platelet storage risk
- TACO vs TRALI
- Delayed reactions and prevention strategies

BOARD-EXAM THEMES

- AHTR root cause (misidentification)
- TACO vs TRALI discriminators (BNP, BP, diuretic response)
- Irradiation for TA-GVHD prevention

SOURCE MATERIALS IN THE REPOSITORY

- TMH S1 C H A P T E R 5 9 Febrile Non Hemolytic Transfusion Reactions (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- TMH S1 C H A P T E R 6 5 Septic Transfusion Reactions (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- TMH S1 C H A P T E R 6 1 Acute Hemolytic Transfusion Reactions (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- Trans Transfusion reactions pathology residents lecture (*Transfusion Medicine / Dr Otrrock lectures*)
- TMH S1 C H A P T E R 6 0 Allergic Transfusion Reactions (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- Harmening 20 Autoimmune Hemolytic Anemia (*Transfusion Medicine / Harmening*)

9. Component Therapy and Massive Transfusion

45–60 min · deck: *Trans_04.pptx*

LEARNING OBJECTIVES

- Select the correct component for the clinical indication.
- State evidence-based thresholds for red cells and platelets.
- Describe plasma, cryoprecipitate, and fibrinogen replacement.
- Outline a massive transfusion protocol.
- Recognize complications of large-volume transfusion.

KEY TOPICS

- RBC, platelet, plasma, cryoprecipitate indications
- Restrictive transfusion thresholds
- Fibrinogen replacement and cryoprecipitate
- Massive transfusion protocols and ratios
- Citrate toxicity, hypothermia, electrolyte shifts
- Viscoelastic-guided resuscitation

BOARD-EXAM THEMES

- Restrictive vs liberal RBC thresholds
- Platelet dose and refractoriness
- Citrate effect: ionized calcium in massive transfusion

SOURCE MATERIALS IN THE REPOSITORY

- Blood Components Alyssa Vitale (*Transfusion Medicine*)
- Apheresis Book chap06 Automated Donations Plasma Red Cells and Multicomponent (*Transfusion Medicine / Apheresis Principles and Practice, 3rd Edition*)
- TMH S1 C H A P T E R 3 4 Cryoprecipitate and Fibrinogen Concentrates (*Transfusion Medicine / Transfusion Medicine and Hemostasis (Book) - Section 1 Transfusion*)
- Thromboelastography (*Transfusion Medicine*)
- Thromboelastography 2 (*Transfusion Medicine*)
- Inherited platelet disorders JAS (*Transfusion Medicine*)

10. Special Topics in Blood Bank

45–60 min · deck: *Trans_05.pptx*

LEARNING OBJECTIVES

- Describe HDFN pathophysiology and prevention.
- State neonatal transfusion considerations.
- Explain Rh immune globulin use and dosing logic.
- Indicate irradiated and CMV-safe products.
- Introduce therapeutic apheresis.

KEY TOPICS

- Hemolytic disease of the fetus and newborn
- Neonatal and intrauterine transfusion
- Rh immune globulin and the fetomaternal hemorrhage estimate
- Irradiated, leukoreduced, CMV-safe products
- Apheresis basics and common indications
- IgA deficiency and washed products

BOARD-EXAM THEMES

- Rh immune globulin candidate selection
- Kleihauer-Betke / FMH dose calculation
- Indications for irradiated components

SOURCE MATERIALS IN THE REPOSITORY

- AABB Chap30 Stem Cell Collection by Apheresis (*Transfusion Medicine / AABB Technical Manual*)
- ApheresisBook chap22 hematopoietic stem cell transplantation an overview (*Transfusion Medicine / Apheresis Principles and Practice, 3rd Edition*)
- Therapeutic apheresis lecture (*Transfusion Medicine / Dr. Eisenbrey Lectures*)
- Transfusion Support for Hematopoietic Stem Cell Transplant Recipients (*Transfusion Medicine / Dr. Eisenbrey Lectures*)
- Harmening 14 Apheresis (*Transfusion Medicine / Harmening*)
- AABB Chap26 Therapeutic Apheresis (*Transfusion Medicine / AABB Technical Manual*)

Block 3 · Clinical Chemistry

5 lectures

11. Analytic Techniques and Preanalytic / Analytic Error

45–60 min · deck: *Chem_01.pptx*

LEARNING OBJECTIVES

- Describe spectrophotometry, immunoassays, and mass spectrometry at a working level.
- Recognize the effects of hemolysis, lipemia, and icterus.
- Identify common immunoassay interferences.
- Distinguish preanalytic, analytic, and postanalytic error.
- Apply the concept of interference to result interpretation.

KEY TOPICS

- Spectrophotometry and Beer–Lambert basics
- Competitive vs sandwich immunoassays
- Chromatography and LC-MS/MS overview
- Hemolysis, lipemia, icterus indices
- Hook effect, heterophile and biotin interference
- Error classification across the testing cycle

BOARD-EXAM THEMES

- Hook (prozone) effect
- Biotin and heterophile interference
- Hemolysis effect on potassium and LDH

SOURCE MATERIALS IN THE REPOSITORY

- Chem Preanalytic Variation July (*Chemistry / Automation*)
- Chem Preanalytic Variation final cf 082017 (*Chemistry / 2017 Chem Rotation - Year 1*)
- Tubes Hemolysis WC VS (*Chemistry / Automation*)
- Centaur instrument overview (*Chemistry / Chem_y1_2*)
- Chem Preanalytical Variation (*Chemistry / 2019 Chem Rotation - Year 1*)
- Chem Preanalytical Variation (*Chemistry / 2024 Chem Rotation - Year 1*)

12. Electrolytes, Osmolality, and Acid–Base Interpretation

45–60 min · deck: *Chem_02.pptx*

LEARNING OBJECTIVES

- Approach hypo- and hypernatremia with tonicity in mind.
- Calculate and interpret the anion gap and osmolal gap.
- Build a stepwise acid–base interpretation.
- Recognize mixed acid–base disorders.
- Identify pseudohyponatremia.

KEY TOPICS

- Sodium disorders and tonicity
- Anion gap and delta-delta

- Osmolal gap and toxic alcohols
- Stepwise acid–base approach
- Compensation rules and mixed disorders
- Pseudohyponatremia (indirect ISE)

BOARD-EXAM THEMES

- High anion gap metabolic acidosis differential
- Osmolal gap in toxic alcohol ingestion
- Pseudohyponatremia mechanism

SOURCE MATERIALS IN THE REPOSITORY

- Chem Osmolality (*Chemistry / 2021 Chem Rotation - Year 2*)
- Chem Osmolality (*Chemistry / 2019 Chem Rotation - Year 1*)
- Chem Osmolality (*Chemistry / 2022 Chem Rotation - Year 1*)
- Chem Arterial Blood Gases (*Chemistry / 2019 Chem Rotation - Year 2*)

13. Renal, Liver, and Cardiac Biomarkers

45–60 min · deck: *Chem_03.pptx*

LEARNING OBJECTIVES

- Interpret creatinine, eGFR, and cystatin C concepts.
- Read liver function test patterns.
- Apply troponin kinetics to acute coronary evaluation.
- Use natriuretic peptides appropriately.
- Recognize common biomarker misuse.

KEY TOPICS

- Creatinine, eGFR equations, cystatin C
- Hepatocellular vs cholestatic LFT patterns
- High-sensitivity troponin kinetics
- BNP / NT-proBNP interpretation
- Biomarker misuse and overordering
- Macro-analytes (macro-CK, macroprolactin)

BOARD-EXAM THEMES

- Hepatocellular vs cholestatic injury
- High-sensitivity troponin delta interpretation
- Cystatin C vs creatinine eGFR

SOURCE MATERIALS IN THE REPOSITORY

- Chem Cardiac Markers May (*Chemistry / Chem_y1_2*)
- Chem Cardiac (*Chemistry / 2026 Chem Rotation Year 1*)
- Chem Cardiac (*Chemistry / 2019 Chem Rotation - Year 2*)
- Chem Cardiac (*Chemistry / 2021 Chem Rotation - Year 2*)
- Chem Cardiac (*Chemistry / 2025 Chem Rotation - Year 1*)
- Chem Cardiac (*Chemistry / 2019 Chem Rotation - Year 1*)

14. Endocrine and Protein Testing

45–60 min · deck: *Chem_04.pptx*

LEARNING OBJECTIVES

- Interpret thyroid function tests and recognize interferences.
- Apply caveats to cortisol and ACTH testing.
- Read SPEP, UPEP, and immunofixation patterns.
- Use serum free light chains in monoclonal gammopathy.
- Recognize assay interferences in endocrine testing.

KEY TOPICS

- TSH-reflex strategy and thyroid interferences
- Cortisol / ACTH dynamic testing caveats
- SPEP / UPEP / IFE pattern recognition
- Serum free light chains and ratios
- Monoclonal gammopathy workup (IMWG)
- Biotin, macro-TSH, and hook effects

BOARD-EXAM THEMES

- SPEP M-spike interpretation and reflex IFE
- Free light chain ratio in monoclonal gammopathy
- Thyroid assay interference (biotin, heterophile)

SOURCE MATERIALS IN THE REPOSITORY

- Chem Protein electrophoresis Keren (*Chemistry*)
- Chem Serum protein electrophoresis Keren (*Chemistry*)
- Protein Electrophoresis Clinical Diagnosis Keren Book (*Chemistry*)
- Vitamin D JS (*Chemistry*)
- Chem Diabetes Kidney (*Chemistry*)
- SPE and Ig discrepancy JS (*Chemistry*)

15. Toxicology, TDM, and Point-of-Care Testing

45–60 min · deck: *Chem_05.pptx*

LEARNING OBJECTIVES

- Distinguish screening immunoassays from confirmatory methods.
- Apply therapeutic drug monitoring principles.
- Recognize drug-screen cross-reactivity and pitfalls.
- State POCT governance and quality requirements.
- Outline method validation essentials for new assays.

KEY TOPICS

- Drug-of-abuse screen vs confirmation (LC-MS/MS, GC-MS)
- TDM: trough/peak timing and steady state
- Cross-reactivity and false positives/negatives
- POCT oversight, connectivity, and competency

- Validation: accuracy, precision, linearity, reportable range
- Reference intervals and critical values

BOARD-EXAM THEMES

- Immunoassay screen cross-reactivity
- TDM sampling timing
- Confirmatory testing rationale

SOURCE MATERIALS IN THE REPOSITORY

- Chem Toxicology Pain Management (*Chemistry / 2018 Chem Rotation - Year 1*)
- Chem GWM Drugs of abuse and pain management (*Chemistry / Morning Conferences*)
- Chem Toxicology Operations GWM (*Chemistry*)
- Chem TDM problem solving gwm (*Chemistry / Morning Conferences*)
- Final Opioid chemistry may 24 GWM edits (1) (*Chemistry / Resident quality projects*)
- Chem Toxicology operations GWM (*Chemistry*)

Block 4 · Microbiology

5 lectures

16. Specimen Selection, Collection, and Rejection Criteria

45–60 min · deck: *Micro_01.pptx*

LEARNING OBJECTIVES

- Match specimen type and transport to the suspected pathogen.
- Apply rejection criteria and explain their rationale.
- Recognize how preanalytic issues degrade culture and molecular testing.
- Reduce contamination risk at collection.
- Communicate specimen problems to clinicians.

KEY TOPICS

- Specimen-to-pathogen matching
- Transport media and time/temperature
- Rejection criteria (e.g., saliva sputum, dry swab)
- Contamination and normal flora
- Anaerobic collection essentials
- Preanalytics for molecular assays

BOARD-EXAM THEMES

- Sputum adequacy (squamous cells)
- Urine culture thresholds and contamination
- Swab vs tissue/aspirate for anaerobes

SOURCE MATERIALS IN THE REPOSITORY

- IntroClinMicroLab ID Fellows Includes core micro (*Microbiology*)
- Chem Preanalytic Variation July (*Chemistry / Chem_y1_2*)
- Chem Preanalytic Variation final cf 082017 (*Chemistry / 2017 Chem Rotation - Year 1*)
- Chem Preanalytical Variation (*Chemistry / 2019 Chem Rotation - Year 1*)
- Chem Preanalytical Variation (*Chemistry / 2024 Chem Rotation - Year 1*)
- AABB Chap06 Whole Blood Collection (*Transfusion Medicine / AABB Technical Manual*)

17. Gram Stain, Culture Workflows, and Organism Identification

45–60 min · deck: *Micro_02.pptx*

LEARNING OBJECTIVES

- Interpret Gram stains and link morphology to likely organisms.
- Follow bench logic from culture to identification.
- Describe rapid identification methods (MALDI-TOF).
- Recognize key colony morphologies.
- Communicate urgent stain and culture findings.

KEY TOPICS

- Gram-stain morphology and reporting
- Primary and selective media

- Colony morphology cues
- Biochemical and rapid identification
- MALDI-TOF mass spectrometry
- Critical-value microbiology communication

BOARD-EXAM THEMES

- Gram-positive cocci in clusters vs chains
- MALDI-TOF principle and limits
- Catalase / coagulase logic

SOURCE MATERIALS IN THE REPOSITORY

- Serology (*Microbiology*)
- Atlas Yeasts IDGuide YEAST (*Microbiology / Mycology*)
- stool culture (*Microbiology*)
- Program improvement meeting (*Microbiology*)
- Board Mycobacteriology Leroy (*Microbiology / Board Review*)
- AbIdentification revision2 (*Transfusion Medicine / Dr. Eisenbrey Lectures*)

18. Antimicrobial Susceptibility Testing and Reporting

45–60 min · deck: *Micro_03.pptx*

LEARNING OBJECTIVES

- Define MIC and interpret breakpoints.
- Compare AST methods (disk diffusion, broth microdilution, gradient).
- Apply cascade and selective reporting.
- Recognize key resistance mechanisms.
- Connect AST to antimicrobial stewardship.

KEY TOPICS

- MIC, breakpoints, and interpretive categories
- Disk diffusion, broth microdilution, gradient strips
- Cascade / selective reporting
- MRSA, ESBL, carbapenemases, VRE
- Intrinsic vs acquired resistance
- Antibigram and stewardship interface

BOARD-EXAM THEMES

- ESBL and carbapenemase detection
- Cascade reporting rationale
- Disk diffusion vs MIC method choice

SOURCE MATERIALS IN THE REPOSITORY

- Micro Abx Res Mech ID Fellows Part 2 (*Microbiology / New abx resistance lectures*)
- Micro Abx Res Mech ID Fellows Part 1 (*Microbiology / New abx resistance lectures*)
- Beta lactam antibiotic resistance BK (*Microbiology*)
- Micro RISE review Abx and Mycology (*Microbiology*)

- Micro Abx Res Mech ID Fellows Part 1 (*Microbiology*)
- Westgard Rules (*Lab Management*)

19. Rapid Diagnostics and Molecular Microbiology

45–60 min · deck: *Micro_04.pptx*

LEARNING OBJECTIVES

- Describe syndromic panels and nucleic acid amplification tests.
- Match rapid tests to the right clinical use case.
- Build a framework for false positives and false negatives.
- State the limitations of molecular detection.
- Integrate rapid results with stewardship.

KEY TOPICS

- NAAT principles and multiplex panels
- Respiratory, GI, meningitis/encephalitis panels
- Detection vs viability vs colonization
- Cycle-threshold caveats
- Result interpretation and reflex testing
- Turnaround time and stewardship impact

BOARD-EXAM THEMES

- Detection of nonviable organisms after treatment
- Multiplex panel false positives
- NAAT vs culture for specific pathogens

SOURCE MATERIALS IN THE REPOSITORY

- Atlas INTRO TO CLINICAL MYCOLOGY (*Microbiology / Mycology*)
- Atlas Yeasts IDGuide YEAST (*Microbiology / Mycology*)
- Intro To Mycologyv2 (*Microbiology*)
- RISE review Mycology (*Microbiology*)
- Micro RISE review Mycology (*Microbiology*)
- Micro RISE review Abx and Mycology (*Microbiology*)

20. Blood Cultures, Sepsis, and Infection-Control Pearls

45–60 min · deck: *Micro_05.pptx*

LEARNING OBJECTIVES

- Interpret blood cultures and distinguish contamination from bacteremia.
- Optimize blood-culture collection (volume, sets, timing).
- Recognize organisms that signal true infection vs skin flora.
- Connect the laboratory to sepsis management.
- Describe the epidemiology / infection-prevention interface.

KEY TOPICS

- Blood-culture volume and number of sets
- Contaminant vs true pathogen clues

- Time-to-positivity and differential
- Rapid identification from positive bottles
- Sepsis bundles and laboratory turnaround
- Outbreak detection and infection control

BOARD-EXAM THEMES

- Single-bottle CoNS: contaminant vs catheter infection
- Blood-culture volume effect on yield
- Time-to-positivity interpretation

SOURCE MATERIALS IN THE REPOSITORY

- Laboratory Safety, Infection Control and Hospital Acquired *(Microbiology)*

Block 5 · Immunology & Serology

4 lectures

21. Immunoassay Principles and Test Performance

45–60 min · deck: *Immuno_01.pptx*

LEARNING OBJECTIVES

- Explain sensitivity, specificity, and predictive values.
- Show how prevalence drives PPV and NPV.
- Recognize heterophile and other immunoassay interferences.
- Apply reflex and confirmatory testing logic.
- Interpret likelihood ratios at a working level.

KEY TOPICS

- Sensitivity / specificity / PPV / NPV
- Prevalence and predictive value
- Heterophile antibody and biotin interference
- Reflex and confirmatory algorithms
- Likelihood ratios
- Gray-zone and equivocal results

BOARD-EXAM THEMES

- Effect of prevalence on PPV
- Heterophile interference recognition
- Screen-then-confirm reasoning

SOURCE MATERIALS IN THE REPOSITORY

- Serology (*Microbiology*)
- Chem Cook Endocrine Immunoassay 1 (*Chemistry*)
- Chem Cook Immunoassay Principles AMC (*Chemistry / Morning Conferences*)
- Chem Introduction to Immunoassays 2 detection methods (*Chemistry / 2017 Chem Rotation - Year 1*)
- Chem Cook Endocrine Immunoassay 1 (*Chemistry / 2026 Chem Rotation Year 1*)
- Chem Cook Endocrine Immunoassay 2 (*Chemistry / 2026 Chem Rotation Year 1*)

22. Autoimmune Serology

45–60 min · deck: *Immuno_02.pptx*

LEARNING OBJECTIVES

- Interpret ANA patterns and titers.
- Use ENA and specific antibodies appropriately.
- Apply ANCA testing in vasculitis.
- Interpret antiphospholipid and complement testing.
- Recognize common overuse patterns.

KEY TOPICS

- ANA by IFA: patterns and titer
- ENA panel and disease associations

- ANCA (c-/p-ANCA, PR3, MPO)
- Antiphospholipid antibody testing
- Complement (C3, C4, CH50)
- Test overuse and pretest probability

BOARD-EXAM THEMES

- ANA pattern–disease associations
- PR3 vs MPO ANCA
- Antiphospholipid laboratory criteria

SOURCE MATERIALS IN THE REPOSITORY

- Analytic Tests in DM AS (*Chemistry*)
- Chem Preatalytic Variation July (*Chemistry / Chem_y1_2*)
- Warm AutoImmune Hemolytic Anemia HM 02262020 (*Hematopathology*)
- Managment of subclinical hypothyroidism in elderly (*Chemistry*)
- Quality expo ppt VF (*Chemistry / Resident quality projects*)
- HFH Poster 4'x6' ACLPS (1) (*Chemistry / Resident quality projects*)

23. Infectious Disease Serology

45–60 min · deck: *Immuno_03.pptx*

LEARNING OBJECTIVES

- Distinguish acute from remote infection serologically.
- Account for window periods.
- Recognize IgM pitfalls.
- Apply confirmatory algorithms (e.g., hepatitis, HIV, syphilis).
- Interpret avidity and seroconversion.

KEY TOPICS

- IgM/IgG kinetics and window periods
- Hepatitis B serologic panel
- HIV 4th-generation algorithm
- Syphilis treponemal/non-treponemal testing
- IgM false positives
- Avidity and paired sera

BOARD-EXAM THEMES

- Hepatitis B panel interpretation
- HIV testing algorithm
- Treponemal vs non-treponemal sequence

SOURCE MATERIALS IN THE REPOSITORY

- Serology (*Microbiology*)
- Micro HIV Hira Qadir (*Microbiology*)
- Micro CP presentation Dec Hepatitis Mehrnoosh (*Microbiology*)
- HepBSerologicChart (*Chemistry*)

- Infectious Disease Testing (*Transfusion Medicine / Dr. Eisenbrey Lectures*)
- OnCall CP on call conference HIV Hira Qadir (*CP On-Call Presentations / 2025*)

24. Allergy, Immunodeficiency, and Protein-Based Immune Testing

45–60 min · deck: *Immuno_04.pptx*

LEARNING OBJECTIVES

- Interpret quantitative immunoglobulins.
- Correlate SPEP with immune findings.
- Apply specific IgE and tryptase testing.
- Outline a targeted immunodeficiency workup.
- Use vaccine-response testing appropriately.

KEY TOPICS

- Quantitative IgG/IgA/IgM/IgE
- SPEP correlation and hypogammaglobulinemia
- Specific IgE and tryptase
- Complement and functional assays
- Vaccine-response (pneumococcal) testing
- When to refer for further immune workup

BOARD-EXAM THEMES

- IgA deficiency and transfusion implications
- Tryptase in anaphylaxis
- Vaccine-response interpretation

SOURCE MATERIALS IN THE REPOSITORY

- hematology basic investigations on call CP role (*Hematopathology*)
- Reportable Diseases Michigan by Pathogen 478489 7 (*Microbiology*)
- Reportable Diseases Michigan by Condition 478488 7 (*Microbiology*)
- Blood group antigens pathology Zaher Otrock HFH (*Transfusion Medicine / Dr Otrock lectures*)
- OnCall CP on call conference hematology basic investigations OZB (*CP On-Call Presentations / 2025*)
- LabMgmt Jones Safety OSHA Latex Allergy quickfacts (*Lab Management / Dr Jones*)

Block 6 · Molecular & Cytogenetics

5 lectures

25. Molecular Methods for Residents

45–60 min · deck: *Molecular_01.pptx*

LEARNING OBJECTIVES

- Describe PCR, RT-PCR, FISH, karyotype, and NGS at a working level.
- Select the right platform for the clinical question.
- Compare turnaround, sensitivity, and breadth across methods.
- Recognize the limits of each technique.
- Interpret basic molecular nomenclature.

KEY TOPICS

- PCR and quantitative / RT-PCR
- FISH and karyotype
- Next-generation sequencing overview
- Platform selection by clinical question
- Analytic sensitivity vs breadth
- HGVS nomenclature basics

BOARD-EXAM THEMES

- FISH vs karyotype vs NGS selection
- Detecting low-level variants
- RT-PCR for fusion transcripts

SOURCE MATERIALS IN THE REPOSITORY

- Techniques in Molecular Pathology Part 1 (Kis) (*Molecular Pathology*)
- Mol Techniques in Molecular Pathology Part 2 (Kis) (*Molecular Pathology*)
- Molecular2 102716 (*Molecular Pathology*)
- Cyto Resident Talk Day One (*Cytogenetics*)
- Cyto Resident Talk Day Two (*Cytogenetics*)
- Cytogenomics – Oncology cases (*Cytogenetics*)

26. Variant Interpretation and Reporting

45–60 min · deck: *Molecular_02.pptx*

LEARNING OBJECTIVES

- Apply variant tiering and pathogenicity frameworks.
- Use standard nomenclature.
- Distinguish somatic from germline interpretation.
- Write clinically actionable reports.
- Recognize variants of uncertain significance and their handling.

KEY TOPICS

- ACMG/AMP and somatic tiering frameworks
- HGVS nomenclature

- Germline vs somatic interpretation
- Actionability and therapy implications
- VUS handling and reanalysis
- Incidental / secondary findings

BOARD-EXAM THEMES

- Tier I vs Tier II somatic variants
- Pathogenic vs VUS classification
- Germline incidental findings

SOURCE MATERIALS IN THE REPOSITORY

- Chem HbA1c Variants (*Chemistry / Chem_y1_2*)
- Chem Critical Value Reporting (*Chemistry / 2025 Chem Rotation - Year 1*)

27. Molecular Testing in Hematologic Malignancy

45–60 min · deck: *Molecular_03.pptx*

LEARNING OBJECTIVES

- Apply fusion and mutation testing in leukemia/lymphoma.
- Use clonality assays appropriately.
- Explain minimal/measurable residual disease (MRD).
- Integrate molecular results with morphology and flow.
- Recognize prognostic and predictive markers.

KEY TOPICS

- BCR-ABL1 and other fusions
- Clonality (IGH/TCR rearrangement)
- MRD concepts and methods
- Integration with morphology and flow
- Prognostic markers (FLT3, NPM1, etc.)
- Sample adequacy for molecular testing

BOARD-EXAM THEMES

- BCR-ABL1 monitoring and MRD
- Clonality testing interpretation
- Integrated diagnosis in acute leukemia

SOURCE MATERIALS IN THE REPOSITORY

- FusionPlex® Pan Solid Tumor FINAL (*Molecular Pathology / Grand Rounds*)
- Mol ELR MDSS Guide V3 Nov (*Molecular Pathology / Informatics*)
- Molecular Toxonomy of MDS (*Molecular Pathology / Molecular Journal club*)
- Molecular2 102716 (*Molecular Pathology*)
- Thyroid Tumor Molecular Genetics (*Molecular Pathology*)
- Mol 3 Network final final 08 (*Molecular Pathology / Informatics*)

28. Solid Tumor and Germline Testing Fundamentals

45–60 min · deck: *Molecular_04.pptx*

LEARNING OBJECTIVES

- Describe companion diagnostics and targeted therapy markers.
- Outline hereditary cancer testing basics.
- Address tissue adequacy and preanalytic concerns.
- Distinguish tumor profiling from germline testing.
- Introduce liquid biopsy concepts.

KEY TOPICS

- Companion diagnostics and targeted therapy
- Hereditary cancer syndromes
- Tissue adequacy, fixation, and tumor fraction
- Tumor profiling vs germline testing
- Liquid biopsy / circulating tumor DNA
- Microsatellite instability and TMB

BOARD-EXAM THEMES

- Companion diagnostic concept
- Tissue adequacy for NGS
- MSI / mismatch-repair testing

SOURCE MATERIALS IN THE REPOSITORY

- Thyroid Tumor Molecular Genetics (*Molecular Pathology*)
- FusionPlex® Pan Solid Tumor FINAL (*Molecular Pathology / Grand Rounds*)
- Heme TB 2 (*Hematopathology / Lymphoma Tumor Board*)
- Heme TB TB (*Hematopathology / Lymphoma Tumor Board*)
- Heme TB TB (*Hematopathology / Lymphoma Tumor Board*)
- Heme TB TB (*Hematopathology / Lymphoma Tumor Board*)

29. Validation, Quality Control, and Bioinformatics Essentials

45–60 min · deck: *Molecular_05.pptx*

LEARNING OBJECTIVES

- Define assay performance characteristics for molecular tests.
- Apply contamination-control practices.
- Build working awareness of the bioinformatics pipeline.
- Recognize sign-out pitfalls.
- Outline proficiency testing for molecular assays.

KEY TOPICS

- Analytic validation: accuracy, precision, LOD
- Contamination control and amplicon carryover
- Bioinformatics pipeline awareness
- Coverage, depth, and quality metrics
- Sign-out pitfalls and artifact recognition
- Proficiency testing and QC

BOARD-EXAM THEMES

- Limit of detection and validation
- Sequencing artifacts vs true variants
- Coverage / depth interpretation

SOURCE MATERIALS IN THE REPOSITORY

- Micro Verification and Validation (*Microbiology*)
- Chem validation of busulfan (*Chemistry / 2025 Chem Rotation - Year 2*)
- Method Validation intro 092017 ref range 6 (*Chemistry / Chem_rotation_y1_2*)
- Chem Quality control basics intro pg1 (*Chemistry / 2023 Chem Rotation - Year 1*)
- Chem Method Validation intro 092017 ref range 6 (*Chemistry / 2017 Chem Rotation-Year 2*)
- Chem LABORATORY QUALITY CONTROL expanded Feldkamp 012016 final (*Chemistry / 2017 Chem Rotation-Year 2*)

Block 7 · Laboratory Management, Informatics, Quality & Safety

5 lectures

30. Quality Management Systems

45–60 min · deck: *LabMgmt_01.pptx*

LEARNING OBJECTIVES

- Differentiate QC and QA within a quality management system.
- Apply Westgard rules and Levey–Jennings interpretation.
- Use quality indicators and proficiency testing.
- Explain delta checks and their role.
- Manage occurrences and corrective action.

KEY TOPICS

- QC vs QA vs QMS
- Levey–Jennings and Westgard multirules
- Proficiency testing and peer comparison
- Quality indicators and metrics
- Delta checks
- Occurrence management and CAPA

BOARD-EXAM THEMES

- Westgard rule violations and actions
- Random vs systematic error patterns
- Delta-check triggers

SOURCE MATERIALS IN THE REPOSITORY

- LabMgmt Jones Quality Proficiency Testing Failure Evaluation 12 (*Lab Management / Dr Jones*)
- Jones Quality Laboratory Quality (*Lab Management / Dr Jones*)
- Jones Quality IHC Quality intro 1 (*Lab Management / Dr Jones*)
- Jones Quality IHC Quality caring 1 (*Lab Management / Dr Jones*)
- Jones Quality IHC Quality variation 1 (*Lab Management / Dr Jones*)
- Jones Quality ContinuousQualityImprovement 1 (*Lab Management / Dr Jones*)

31. Regulatory and Accreditation Essentials

45–60 min · deck: *LabMgmt_02.pptx*

LEARNING OBJECTIVES

- Describe CAP and CLIA frameworks at a working level.
- Prepare for inspection readiness.
- Apply competency-assessment requirements.
- Maintain document control.
- Recognize test-complexity categories.

KEY TOPICS

- CLIA complexity categories
- CAP accreditation and checklists

- Inspection readiness
- Competency assessment (six elements)
- Document control and version management
- Personnel and quality requirements

BOARD-EXAM THEMES

- CLIA waived vs moderate vs high complexity
- Six elements of competency assessment
- Proficiency-testing regulatory requirements

SOURCE MATERIALS IN THE REPOSITORY

- LabMgmt Jones Compliance CAP Compliance Planning course FINAL DRAFT DECK tb pptx (*Lab Management / Dr Jones*)
- LabMgmt Jones Compliance CAP Compliance Planning DRAFT QUIZ tb (*Lab Management / Dr Jones*)
- Info 3 HL7 Data Formats harrison jmt modifications (*Pathology Informatics*)
- LabMgmt Jones MedicalDirector CAP Lecture Medical Director FINAL DRAFT DECK tb (*Lab Management / Dr Jones*)
- LabMgmt Jones CLIA CLIA 88 (*Lab Management / Dr Jones*)
- LabMgmt Jones CLIA CLIA 88 highlighted (*Lab Management / Dr Jones*)

32. Patient Safety and Error Investigation

45–60 min · deck: *LabMgmt_03.pptx*

LEARNING OBJECTIVES

- Conduct a structured root-cause analysis.
- Distinguish near misses from events.
- Investigate transfusion-safety events.
- Apply systems thinking to laboratory error.
- Build a just-culture approach.

KEY TOPICS

- Root-cause analysis methods
- Near miss vs adverse event
- Transfusion-safety events and haemovigilance
- Human factors and systems design
- Just culture
- Corrective and preventive action

BOARD-EXAM THEMES

- RCA vs blame in laboratory error
- Near-miss reporting value
- Sentinel-event response

SOURCE MATERIALS IN THE REPOSITORY

- LabMgmt Jones Risk Medicolegal aspect of error in pathology (*Lab Management / Dr Jones*)
- LabMgmt Deviation and Daily Management GWM (*Lab Management*)

- LabMgmt Deviation and Daily Management GWM (*Lab Management*)
- LabMgmt Deviation and Daily Management Cook (*Lab Management*)
- Jones Safety Laboratory Safety (*Lab Management / Dr Jones*)
- LabMgmt Jones Risk Risk Management in Pathology Laboratory Medicine (*Lab Management / Dr Jones*)

33. Laboratory Informatics and LIS Fundamentals

45–60 min · deck: *LabMgmt_04.pptx*

LEARNING OBJECTIVES

- Describe LIS architecture and result routing.
- Explain middleware and autoverification.
- Recognize interface issues and data-integrity risks.
- Apply standard vocabularies (LOINC, SNOMED).
- Outline downtime and contingency planning.

KEY TOPICS

- LIS architecture and workflow
- Middleware and autoverification rules
- Interfaces (HL7) and failure modes
- Data integrity and result routing
- LOINC / SNOMED coding
- Downtime procedures

BOARD-EXAM THEMES

- Autoverification rule design
- HL7 interface failure recognition
- LOINC vs SNOMED use

SOURCE MATERIALS IN THE REPOSITORY

- Info 2 Network final final (*Pathology Informatics*)
- Info 3 HL7 Data Formats harrison jmt modifications (*Pathology Informatics*)
- Info 5 Cybersecurity for Pathology residents HFH ozb (*Pathology Informatics*)
- Info 1 PC hardware and software basic (*Pathology Informatics*)
- Info 1 Pathology Informatics talk final rev (*Pathology Informatics*)
- LabMgmt Jones CLIA CLIA Waived test list (*Lab Management / Dr Jones*)

34. Utilization Management, Cost Awareness, and Leadership

45–60 min · deck: *LabMgmt_05.pptx*

LEARNING OBJECTIVES

- Apply test-utilization and stewardship principles.
- Use formulary thinking for send-outs.
- Practice consultative pathology.
- Outline a quality-improvement project.
- Develop laboratory leadership skills.

KEY TOPICS

- Test utilization and overuse reduction
- Send-out formulary management
- Consultative pathology practice
- QI project design (PDSA)
- Cost awareness and value
- Leadership and change management

BOARD-EXAM THEMES

- Utilization intervention design
- Appropriate send-out gatekeeping
- PDSA cycle in a QI project

SOURCE MATERIALS IN THE REPOSITORY

- LabMgmt Jones Finance CAP Cost Analysis Toolkit (*Lab Management / Dr Jones*)
- LabMgmt Matters Relating to Clinical Testing (*Lab Management*)
- Jones Leadership Leadership (*Lab Management / Dr Jones*)
- LabMgmt Matters Relating to Clinical Testing Cook (*Lab Management*)
- LabMgmt Matters Relating to Clinical Testing Cook (*Lab Management*)
- LabMgmt Matters Relating to Clinical Testing Cook (*Lab Management*)

Yearly Delivery Model

A block-based schedule keeps teaching systematic while allowing repetition and board review. One subspecialty runs at a time in 4–5 weekly sessions; 34 sessions fill most of an academic year and leave room for journal club, unknown conferences, and multidisciplinary case conferences.

Hematology & Coagulation	5	Weeks 1–5
Transfusion Medicine	5	Weeks 6–10
Clinical Chemistry	5	Weeks 11–15
Microbiology	5	Weeks 16–20
Immunology & Serology	4	Weeks 21–24
Molecular & Cytogenetics	5	Weeks 25–29
Laboratory Management, Informatics, Quality & Safety	5	Weeks 30–34
Total	34	≈ one academic year

Each session runs 45–60 minutes including 10–15 minutes of case discussion.

Slide-Deck Design System

All 34 decks share one visual identity — “Clinical Navy” with a distinct accent color per block, so each subspecialty is instantly recognizable. Every deck uses the same slide types: title, learning objectives, roadmap, section dividers, concept cards, original algorithms, comparison tables, teaching cases with answer reveals, board-style questions, key take-home points, and references.

Hematology & Coagulation	Crimson	#C62133
Transfusion Medicine	Magenta	#AD1457
Clinical Chemistry	Blue	#1565C0
Microbiology	Green	#2E7D32
Immunology & Serology	Purple	#6A1B9A
Molecular & Cytogenetics	Teal	#00838F
Laboratory Management, Informatics, Quality & Safety	Slate	#455A64

Notes on Sources and Gaps

The lecture–source crosswalk above was generated by matching each lecture against the indexed repository. Coverage is strongest where the department already teaches heavily: transfusion medicine, hematology/coagulation, chemistry, and laboratory management. Topics with lighter repository coverage — to be developed or supplemented from current standards — include:

- Immunology & serology (no dedicated folder; material is dispersed across chemistry and microbiology).
- Molecular variant interpretation and reporting frameworks (ACMG/AMP tiering).
- Molecular assay validation and bioinformatics essentials.

These gaps are opportunities: building these decks also builds reusable teaching assets the department currently lacks.